

Multiple Choice (10 marks)

1. In Python 3, which of the following function convert a string to an int in python?
 - (a) `int(x [,base])`
 - (b) `long(x [,base] )`
 - (c) `float(x)`
 - (d) `str(x)`
2. Let `l = [1,2,3,4]`. What is `min(l)` in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
3. Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-ofservice (D o S) attack?
 - (a) The goal of the attack
 - (b) The number of computers being attacked
 - (c) The number of computers launching the attack
 - (d) The time period in which the attack occurs
4. In Python 3, what is the output of `print tuple[0]` if `tuple = ( 'abcd', 786 , 2.23, 'john', 70.2 )`?
 - (a) `( 'abcd', 786 , 2.23, 'john', 70.2 )`
 - (b) `abcd`
 - (c) Error
 - (d) None of the above.
5. Which types of functions grow the slowest?
 - (a) $O(N^{1/2})$
 - (b) $O(N^{1/4})$
 - (c) $O(N^{1/N})$
 - (d) $O(N)$

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression 'a' + 'ab' produces an error due to incompatible types.
7. True or False: In Python 3, the function cmp(list) is used to find the item with the maximum value in a list.
8. True or False: Confidentiality is the pillar of cybersecurity that is affected when a user logs in with unauthorized credentials.
9. True or False: A device driver is assigned to a device when it connects to the Internet.
10. True or False: The boolean expression $a[i] == \max \parallel !(\max \neq a[i])$ simplifies to $a[i] \neq \max$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to replace the last element of the list with another list.
12. Write a python function to count the number of lists in a given number of lists.

End of Paper